

View results

Respondent

25

Anonymous

70:45

Time to complete

A. General information about the respondent

1. * Country

Belarus



2. * Name of organization(s) or entity(s) responsible for the preparation of the report

State Committee on Science and Technology of the Republic of Belsrus, Ministry of Education of the Republic of Belarus, Ministry of economy of the Republic of Belarus, National Academy of sciences of Belarus

3. * Website of organization(s) or entity(s) responsible for the preparation of the report

scst.gov.by; edu.gov.by; economy.gov.by; nasb.gov.by

4. * Email address of organization(s) or entity(s) responsible for the preparation of the report

5. Phone number of organization(s) or entity(s) responsible for the preparation of the report

6. Please describe the role/mandate of your organization

The State Committee on Science and Technology of the Republic of Belarus (hereinafter referred to as the SCST) is a republican body of state administration that implements state policy, carries out regulation and management in the areas of scientific, technical and innovative activities, as well as in the area of protection of rights to intellectual property objects and coordinates the activities of other republican bodies of state administration in these areas.

7. * Full name of officially designated contact person(s) who completed this survey

Sergei Siniukovich

8. * Position of officially designated contact person(s) who completed this survey

Deputy Head Department of International S&T policy

9. * Email address of officially designated contact person(s) who completed this survey

10. Full Name(s) of designated official(s) certifying the report

Sergei Siniukovich

11. Brief description of the consultation process established for the preparation of the report

Ministry of Education of the Republic of Belarus, Ministry of economy of the Republic of Belarus, National Academy of sciences of Belarus took part in preparation process

12. * Name of other organization(s) or entit(y)(ies) (including non-governmental) consulted for completing this survey

Ministry of Education of the Republic of Belarus, Ministry of economy of the Republic of Belarus, National Academy of sciences of Belarus

13. * Website of other organization(s) or entit(y)(ies) (including non-governmental) consulted for completing this survey
(Please use commas to separate multiple links, as: link A, link B)

scst.gov.by; edu.gov.by; economy.gov.by; nasb.gov.by

14. * Email address of other organization(s) or entit(y)(ies) (including non-governmental) consulted for completing this survey

15. Sector of other organization(s) or entit(y)(ies) (including non-governmental) consulted for completing this survey

- ☐ Public
- ☐ Private
- ☐ Civil Society
- ☒ official governmental bodies

B. QUESTIONNAIRE FOR MEMBER STATES' REPORTING

ON THE IMPLEMENTATION OF THE 2021 RECOMMENDATION ON OPEN
SCIENCE

Part 1. PROMOTING A COMMON UNDERSTANDING OF OPEN SCIENCE,

ASSOCIATED BENEFITS AND CHALLENGES, AS WELL AS DIVERSE PATHS TO
OPEN SCIENCE

16. 1.1 Has the 2021 UNESCO Recommendation on Open Science been promoted and/or shared with appropriate ministries and institutions as well as affiliated organizations in your country?

- ☐ Yes
- ☒ No

17. 1.1 cont. **If no**, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.

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18. 1.2 Is the 2021 Recommendation on Open Science available in the national language(s) of your country?

☒ Yes

☐ No

19. 1.2 cont. **If yes**, which language(s)?

Russian

20. 1.3 Have there been awareness raising activities on open science, including all its key elements[i], associated benefits and challenges, organized or foreseen to be organized by the end of 2025 in your country by national authorities or entities? (ref.: (i) 16.f, 16.g, 16.h)

[i] Please refer to the elements of open science defined in the 2021 Recommendation, namely : open scientific knowledge (including open access to scientific publications, research data, open educational resources, open-source software and code, open hardware), open science infrastructures, open engagement of societal actors and open dialogue with other knowledge systems.

☒ Yes

☐ No

21. 1.3 cont. **If yes**, please provide details on the activities organized or foreseen and links as relevant.

In this part, scientific conferences, forums, and exhibitions financed from budgetary funds are held in Belarus. Examples of such events include: the annual International Scientific Conference "Problems of forecasting and state regulation of socio-economic development" held at the Scientific Research Economic Institute of the Ministry of Economy of the Republic of Belarus; the annual International Scientific Conference "Youth in Science - 2024" held at the National Academy of Sciences of Belarus; the annual International Forum on Information and Communication Technologies (TIBO), which includes the International Specialized Exhibition;

16.g - Involvement of the private sector in the discussion of ways to expand the scope of application and joint implementation of the principles and priorities of open science. In order to ensure interaction between government agencies and representatives of the private sector, a number of public councils operate in Belarus: Public Advisory (Expert) Council for the Development of Entrepreneurship in the Scientific, Technical and Innovative Spheres under the State Committee on Science and Technology of the Republic of Belarus (Regulations on the Council: <https://www.gknt.gov.by/upload/Deyatelnost/Polozenie%20ob%20OKS.pdf>); Interdepartmental Supervisory Council of Scientific and Educational Networks of Belarus under the National Academy of Sciences of Belarus (composition of the council: <https://nasb.gov.by/rus/docs/sovety/nauch-obr.pdf>); Public Advisory Council for the Development of Entrepreneurship under the Ministry of Education of the Republic of Belarus (Regulations on the Council: <https://edu.gov.by/sistema-obrazovaniya/obshchestvennye-sovety/obshchestvennye-sovety/obschestvenno-konsultativnyj-sovet-po-razvitiyu-predprinimatelstva/>). Active interaction of government bodies with subordinate bodies and organizations, with local government and self-government bodies, with the private sector and the public is carried out in the process of scientific support for the development of the country's program and forecast documents: the National Strategy for Sustainable Development of the Republic of Belarus, the Program for the Socio-Economic Development of the Republic of Belarus. As part of this activity, round tables and working groups are held.

16.h – Creating conditions for open multilateral discussions in order to discuss the benefits of open science and the real and perceived problems associated with it... In order to ensure a multilateral discussion on the problems, prospects for the use of intellectual property objects, the National Intellectual Property Center holds a large number of thematic events (international scientific and practical conferences, international seminars, etc.): annual International Scientific and Practical Conference "Intellectual Property in the Modern World: Challenges of the Time and Development Prospects"; International seminar "IP and SDGs: innovation and creativity for the benefit of common future" (https://www.ncip.by/o-centre/novosti/sobytiya/24.04_mezhdunarodnyy-seminar-is-i-tsur-innovatsii-i-tvorchestvo-na-bлаго-obshchego-budushchego-proshel-v-m/?sphrase_id=111190); Congress of Technology and Innovation Support Centers of the Republic Belarus (https://www.ncip.by/o-centre/novosti/sobytiya/28.11_ii-sezd-tsentrov-podderzhki-tekhnologiy-i-innovatsiy-respubliki-belarus/); International seminar on the issues of valuation of IP objects and their registration in accounting (https://www.ncip.by/o-centre/novosti/sobytiya/14.12_mezhdunarodnyy-seminar-po-voprosam-otsenki-stoimosti-obektov-is-i-postanovki-ikh-na-bukhgalterskiy-u/).

22. 1.4 Have specific actions been undertaken or are planned to be undertaken by end of 2025 in your country to incorporate the values and principles of open science[i], in publicly funded research? (ref.: (i) 16.a, 16.b, 16.c, 16.d, 16.e, 16.h)

[i] The 2021 Recommendation outlines a set of shared values of open science stemming from the rights-based, ethical, epistemological, economic, legal, political, social, multi-stakeholder and technological implications of opening science to society and broadening the principles of openness to the whole cycle of scientific research. It also provides a set of guiding principles, as a framework for enabling conditions and practices, which uphold the values of open science and can realize the ideals of open science.

Please see the values of open science, namely quality and integrity, collective benefit, equity and fairness, and diversity and inclusiveness.

Please see the guiding principles for open science ,namely transparency, scrutiny, critique and reproducibility; equality of opportunities; responsibility, respect and accountability; collaboration, participation and inclusion; flexibility and sustainability.

☒ Yes

☐ No

23. 1.4 cont. If **yes**, please indicate the actions that are taken to incorporate the specific values and principles. For each case, please provide details and links as relevant.

16.b – Ensuring that publicly funded research is conducted on the basis of open science principles and in accordance with the provisions of this recommendation. Reports on research projects subject to state registration are sent to the Belarusian Institute of 4 System Analysis and Information Support for Science and Technology (BellSA), which in turn maintains a register of them and, upon request from interested parties, provides access to the results of the R&D(T)R. In addition, BellSA publishes a collection of abstracts of unpublished documents to familiarize organizations and specialists in the country with the results of completed R&D(T)R and deposited manuscripts. Also, a prerequisite for the provision of grants by the Belarusian Republican Foundation for Fundamental Research (BRFFR) is the obligation of scientists to make the results of their research available to the public by publishing them in scientific journals with a mandatory indication of financial support from BRFFR.

16.c – Promoting bibliodiversity through diversification of publication formats and media, including in the humanities and social sciences. Examples of open access resources in the Republic of Belarus: Belarusskaya Palichka is the largest and oldest Belarusian electronic library. In addition to Belarusian and foreign works translated into Belarusian, it also contains translations of works by Belarusian writers and poets into other languages (Russian, English, Polish, German), books and texts on literary criticism, history of political science, religious studies, philosophy, etc. The interface is available in Belarusian (<https://knihi.com/>); Belarusian National Educational Internet Portal – full-text electronic teaching aids and teaching aids for various levels of education. The interface is available in Russian (<https://e-vedy.adu.by/>); Gomel State University named after F. Skorina - electronic archive of scientific publications of employees and teachers of Gomel State University named after F. Skorina. Provides access to conference and seminar materials, periodical publications of the university, publications of the GSU library, works of GSU scientists, materials for training. The interface is available in Russian and English (<https://elib.gsu.by/>); BellSA publications - this section of the site presents full texts of issues of the journal "Science and Technology News", registers of registered R&D, R&D, R&D, registration bulletins, issues of the abstract collection "Reports on R&D, R&D, R&D", as well as information and analytical publications of the State Committee on Science and Technology, issues of the publication "Regulatory legal support for scientific and technical activity in the Republic of Belarus", catalogs and conference materials. The interface is available in Russian (<http://www.belisa.org.by/ru/izd/>); national legal Internet portal of the Republic of Belarus - the basis of the information content of the Portal is legal information (texts of legal acts of the Republic of Belarus). This Portal also contains scientific and practical, educational, reference and other information in the field of law and legal informatization. The interface is available in Russian and English (<https://pravo.by/>); national fund of technical regulatory legal acts - contains technical regulatory legal acts in the field of technical regulation and standardization, aviation rules, zoohygienic, veterinary, veterinary-sanitary norms and rules, sanitary norms, rules and hygienic standards, fire safety norms and rules, norms and rules to ensure technical, industrial, nuclear and radiation safety, norms and rules to ensure the safe transportation of dangerous goods, protection and rational use of subsoil, qualification reference books, state classifiers of technical and economic information, forms of state statistical reporting and instructions for filling them out, methods for the formation and calculation of statistical indicators, instructions for organizing and

Part 2. DEVELOPING AN ENABLING POLICY ENVIRONMENT FOR OPEN SCIENCE

24. 2.1 Does your country have a national policy[i], strategy or plan of action on science, technology and innovation (STI)?

[i] National STI policy is most commonly a governmental document developed by governing bodies leading the STI policy design and implementation in a given country, which formulates objectives and guides actions and decision-making processes in the area of STI on the national level.

☒ Yes

☐ No

25. 2.1 cont. **If yes**

- please share a machine-readable PDF file of the policy using this link: <https://tinyurl.com/27p5zct8> designating openscience@unesco.org as the recipient, and provide the following information: *Title of the policy, the authority in charge of development of the policy, entities involved and the process of development, year of adoption, year of last update, key areas/sections of the policy, and links to the webpage if available.*

In the comment box, please indicate the question number for this attachment / document.

After uploading, please return to this page to continue the survey.

Such documents are: 1. Strategy "Science and Technology: 2018 - 2040" (https://nasb.gov.by/congress2/strategy_2018-2040.pdf). The strategy was developed by the National Academy of Sciences of Belarus and approved by the Resolution of the Presidium of the National Academy of Sciences of Belarus dated 26.02.2018 No. 17. 2. State Program of Innovative Development of the Republic of Belarus for 2021 - 2025 (http://www.belisa.org.by/pdf/2022/gpir_2021-2025_22.pdf). Approved by the Decree of the President of the Republic of Belarus dated September 15, 2021 No. 348. The State Program for Innovative Development of the Republic of Belarus for 2021–2025 is the main document ensuring the implementation of the main directions of the state innovation policy. 3. Sections concerning the development of scientific, scientific-technical and innovative spheres are also contained in the National Strategy for Sustainable Development of the Republic of Belarus until 2035 (<https://economy.gov.by/uploads/files/ObsugdaemNPA/NSUR-2035-1.pdf>) and in the Program of Socio-Economic Development of the Republic of Belarus for 2021-2025 (https://president.gov.by/fp/v1/990/document-thumb_33990__original/33990.1629192589.a630b47e0e.pdf). The National Strategy for Sustainable Development of the Republic of Belarus until 2035 was approved by the Minutes of the meeting of the Presidium of the Council of Ministers of the Republic of Belarus dated 04.02.2020 No. 3. The Program of Social and Economic Development of the Republic of Belarus for 2021-2025 was approved by the Decree of the President of the Republic of Belarus dated 29.07.2021 No. 292. The National Strategy for Sustainable Development of the Republic of Belarus until 2035 and the Program of Social and Economic Development of the Republic of Belarus for 2021-2025 were developed by the Ministry of Economy jointly with the Research Economic Institute of the Ministry of Economy of the Republic of Belarus with the involvement of interested government agencies and organizations, local government and self-government bodies, and the public in the discussion. The above state documents are based on the following principles of open science: open access to scientific knowledge; open scientific infrastructure; open participation of social actors; open dialogue with other knowledge systems.

26. 2.1 cont. **If yes:**

- does it include any of the following open science elements:

- ☐ open access to scientific knowledge
- ☐ open science infrastructure
- ☐ open engagement of societal actors
- ☐ open dialogue with other knowledge systems

27. 2.2 In your country, is there a policy, or a set of policies and/or legal framework(s)[i] at the national level that address open science or any of its key elements in line with the definition, values and principles outlined in the 2021 Recommendation on Open Science? (*ref.: (i) 16.a, 16.b, 16.c, 16.d, 16.e, 16.h*)

[i] These can include specific national policies, sets of guidelines, rules, regulations, laws, principles, or directions to govern open science practices.

- ☒ Yes
- ☐ Partly
- ☐ No

28. 2.2 cont. **If yes or partly**, please share a machine-readable PDF file(s) of the policy(ies) using this link: <https://tinyurl.com/27p5zct8> designating openscience@unesco.org as the recipient, and provide the following information for each: *Title of the policy or legal instrument, the authority in charge of development of the policy, entities involved and the process of development, year of adoption, year of last update, key areas/sections of the policy, elements of open science that are addressed, and links to the webpage(s) if available.*

Such documents are: 1. Strategy "Science and Technology: 2018 - 2040" (https://nasb.gov.by/congress2/strategy_2018-2040.pdf). The strategy was developed by the National Academy of Sciences of Belarus and approved by the Resolution of the Presidium of the National Academy of Sciences of Belarus dated 26.02.2018 No. 17. 2. State Program of Innovative Development of the Republic of Belarus for 2021 - 2025 (http://www.belisa.org.by/pdf/2022/gpir_2021-2025_22.pdf). Approved by the Decree of the President of the Republic of Belarus dated September 15, 2021 No. 348. The State Program for Innovative Development of the Republic of Belarus for 2021–2025 is the main document ensuring the implementation of the main directions of the state innovation policy. 3. Sections concerning the development of scientific, scientific-technical and innovative spheres are also contained in the National Strategy for Sustainable Development of the Republic of Belarus until 2035 (<https://economy.gov.by/uploads/files/ObsugdaemNPA/NSUR-2035-1.pdf>) and in the Program of Socio-Economic Development of the Republic of Belarus for 2021-2025 (https://president.gov.by/fp/v1/990/document-thumb__33990__original/33990.1629192589.a630b47e0e.pdf). The National Strategy for Sustainable Development of the Republic of Belarus until 2035 was approved by the Minutes of the meeting of the Presidium of the Council of Ministers of the Republic of Belarus dated 04.02.2020 No. 3. The Program of Social and Economic Development of the Republic of Belarus for 2021-2025 was approved by the Decree of the President of the Republic of Belarus dated 29.07.2021 No. 292. The National Strategy for Sustainable Development of the Republic of Belarus until 2035 and the Program of Social and Economic Development of the Republic of Belarus for 2021-2025 were developed by the Ministry of Economy jointly with the Research Economic Institute of the Ministry of Economy of the Republic of Belarus with the involvement of interested government agencies and organizations, local government and self-government bodies, and the public in the discussion. The above state documents are based on the following principles of open science: open access to scientific knowledge; open scientific infrastructure; open participation of social actors; open dialogue with other knowledge systems.

29. 2.3 In your country, are there specific policy instruments[i] that aim to promote open science?

[i] Policy instruments include programmes, methods and mechanisms of technical and operational nature, required to solve the issues set by the policy, with focus on the target beneficiaries, resources, indicators, and set of goals for delivering products (short term), results (medium-term), and impacts (long term).



Yes



Under development



No

30. 2.3 cont. **If yes or under development**, please provide details and links as relevant, including for each instrument: *Title of the instrument, its objective, the authority in charge, allocated budget, elements of open science included.*

The policy instruments aimed at developing open science in Belarus include: 1. Strategy "Science and Technology: 2018 - 2040" (https://nasb.gov.by/congress2/strategy_2018-2040.pdf). Responsible body: National Academy of Sciences of Belarus. The Strategy defines: key features of the future intellectual economy and new contours of its production system; basic foundations, goals, objectives and priorities for the development of the scientific and technological sphere; the main directions of state policy in science and innovation and instruments for stimulating the scientific and technological development of the national economy for the period up to 2040; expected results of the Strategy implementation. It is based on the following principles of open science: open access to scientific knowledge; open scientific infrastructure; open participation of social actors; open dialogue with other knowledge systems. 2. State Program of Innovative Development of the Republic of Belarus for 2021-2025 (http://www.belisa.org.by/pdf/2022/gpir_2021-2025_22.pdf). Responsible body: State Committee on Science and Technology of the Republic of Belarus. The state program contributes to the implementation of the priorities of the socio-economic development of the Republic of Belarus for 2021-2025 in the field of effective investments and accelerated development of innovative sectors of the economy, the main directions of the state innovation policy, priority areas of scientific, scientific-technical and innovative activities for 2021-2025 and the further development of the national innovation system. Is based on the following principles of open science: open access to scientific knowledge; open scientific infrastructure; open participation of social actors; open dialogue with other knowledge systems. 3. State Program "Science-Intensive Technologies and Equipment" for 2021-2025 (https://nasb.gov.by/rus/activity/issledovaniya/doc/245_2.pdf). Responsible body: National Academy of Sciences of Belarus. The goal of the State Program is to form a knowledge-intensive economy by ensuring scientific, economic and geopolitical interests of the republic, national bioresource and environmental security, technological development of industries in order to achieve competitive advantages. It is based on the following principles of open science: open access to scientific knowledge; open scientific infrastructure; open participation of social actors; open dialogue with other knowledge systems. 4. State Program "Scientific and Innovative Activities of the National Academy of Sciences of Belarus" for 2021-2025 (https://nasb.gov.by/rus/activity/issledovaniya/doc/238_2.pdf). Responsible body: National Academy of Sciences of Belarus. The goal of the State Program is to develop scientific and innovative activities and improve the infrastructure of the NAS of Belarus. It is based on the following principles of open science: open access to scientific knowledge; open scientific infrastructure; open participation of social actors; open dialogue with other knowledge systems. 5. Within the framework of the Union State of Belarus and Russia, a single scientific space is formed by implementing joint scientific programs. Belarusian and Russian scientists jointly develop intelligent electronics, robots and electric motors for cars, laser systems for communication systems and innovative medical products and equipment. The most promising areas are microelectronics, agriculture, medicine and biosafety, social sciences, mechanical engineering, artificial intelligence. In addition, the Resolution of the Supreme State Council of the Union State dated January 29, 2024 No. 2 approved the Strategy for Scientific and Technological Development of the Union State for the period up to 2035 (<https://soyuz.by/projects/dekrety-vysshego-gosudarstvennogo-soveta-soyuznogo-gosudarstva/postanovlenie-ot-29-yanvarya-2024-g-2-o-strategii-nauchno-tehnologicheskogo-razvitiya-soyuznogo-gosudarstva-na-period>

31. 2.4 In your country, is there a national implementation plan, strategy, policy or a roadmap for implementation of open science at the national level in line with the 2021 Recommendation? (ref.: (ii) 17.a, 17.b, 17.c, 17.f, 17.h)



Yes



Under development



No

32. 2.4 cont. **If yes or under development**, please provide details and links as relevant, including the name of ministries and national authorities/entities, as well as affiliated organizations that are involved in the development and implementation of the plan, strategy or the roadmap.

Belarus has developed a regulatory framework that complies with the principles of open science. For example: – Resolution of the Council of Ministers of the Republic of Belarus dated February 9, 2017 No. 110 "On approval of the Model Regulations on the industry laboratory" (<https://pravo.by/document/?guid=12551&p0=C21700110&p1=1>). An industry laboratory is created in order to ensure the implementation of research, development and experimental-technological work, scientific support for innovative projects, pilot testing and the introduction of the results of scientific and scientific-technical activities into production in organizations in a specialized field or industry. – Resolution of the State Committee on Science and Technology of the Republic of Belarus of June 20, 2017 No. 13 "On assigning the status of a center for collective use of unique scientific equipment to organizations (their structural divisions)" (<https://pravo.by/document/?guid=12551&p0=W21732196&p1=1>). In order to improve the efficiency of using scientific equipment necessary for the development of priority areas of scientific and scientific-technical activity, a network of centers for collective use of unique scientific equipment (CSU) is being developed in the Republic of Belarus. 17.b – Coordination of policies, strategies and actions in the field of open science at different levels - from individual institutions to the local and international levels, while respecting the diversity of scientific approaches. In accordance with the Law of the Republic of Belarus of May 5, 1998 №157-3 "On state forecasting and state planning" (<https://etalonline.by/document/?regnum=h19800157>) state forecasting and planning documents are subordinated and coordinated in terms of goals, priorities, objectives, activities, resources, indicators and implementation deadlines. Thus, sections concerning the development of scientific, scientific-technical and innovation spheres are developed on the basis of the Comprehensive Forecast of scientific and technological progress of the Republic of Belarus within the framework of the National Strategy for Sustainable Development of the Republic of Belarus (long-term forecasting document), the Program for the Socio- Economic Development of the Republic of Belarus, state programs, programs for the development of economic sectors (types of economic activity), programs for the socio-economic development of administrative-territorial units of the regional and basic territorial levels (medium-term planning documents). In addition, all draft documents of state forecasting and state planning are necessarily submitted for public professional discussion. 17.c – Comprehensive consideration of gender equality aspects in open science policies, strategies and practices. In the Republic of Belarus, there is no problem of gender inequality in the scientific, scientific-technical and innovation spheres. Thus, as of 2023, the share of women in the total number of researchers was 39% (among candidates of science - 41%, among doctors of science - 22%). 17.f – Expanding the integration of civic and participatory science into open science policies and practices at the national, institutional and funding levels. In this part, in Belarus, in order to ensure interaction between government agencies and representatives of the private sector in the country, a number of public councils operate: Public Advisory (Expert) Council for the Development of Entrepreneurship in the Scientific, Technical and Innovative Spheres under the State Committee on Science and Technology of the Republic of Belarus (Regulations on the Council: <https://www.gknt.gov.by/upload/Deyatelnost/ Polozenie%20ob%20OKS.pdf>); Interdepartmental Supervisory Council of Scientific and Educational Networks of Belarus under the National Academy of Sciences of Belarus (Composition of the Council: <https://nasb.gov.by/rus/docs/sovety/nauch-obr.pdf>); 17.h – Support for responsible practices in assessing and an

33. 2.5 In your country, are there policies and/or strategies that promote open science in line with the 2021 Recommendation at **institutional** level, including in the context of research-performing institutions, universities, scientific unions and associations and learned societies? (ref.: (ii) 17.d, 17.e, 17.g)

- ☒ Yes
- ☐ Under development
- ☐ No

34. 2.5 cont. **If yes**, please provide more information, including the number of policies, name of the institute(s) that has/have developed the policy(s) and elements of open science included in the policy (please consider policies and strategies addressing all elements of open science, including citizen and participatory science, or co-production of knowledge with multiple actors).

To view the elements of open science,

see: <https://unesdoc.unesco.org/ark:/48223/pf0000379949/PDF/379949eng.pdf.multi.nameddest=20/p.nameddest=20/.page=9>

17.d – Encouraging research institutions, primarily those receiving public funds, to apply policies and strategies in the interests of open science. Reports on research projects subject to state registration are sent to the Belarusian Institute of System Analysis and Information Support for Science and Technology (BellSA), which in turn maintains a register of them and, upon request, provides access to the results of R&D. In addition, BellSA publishes a collection of abstracts of unpublished documents in order to familiarize organizations and specialists in the country with the results of completed R&D and deposited manuscripts. Also, a necessary condition for the provision of grants by the Belarusian Republican Foundation for Fundamental Research (BRFFR) is the obligation of scientists to make the results of their research publicly available by publishing them in scientific journals with a mandatory indication of financial support from BRFFR. Scientific institutions and representatives of state organizations publish analytical information on their websites on the results of the conducted research. Examples include: State Scientific Institution "Research Economic Institute of the Ministry of Economy of the Republic of Belarus" - <https://niei.by/ru/nauch-izdaniya-ru/>; National Academy of Sciences of Belarus - <https://nasb.gov.by/rus/activity/razrabotki/>. 17. g - Development of models that ensure joint creation of knowledge with the participation of various entities, and development of principles for the recognition of partnerships that go beyond the scientific community. In this part, in the republic, when developing tasks for state scientific research programs, the National Academy of Sciences of Belarus announces a competition of research works for their inclusion as tasks in state scientific research programs, in which both scientists and researchers from the academic environment, and from university science and independent researchers can take part (<https://nasb.gov.by/rus/news/10514/>).

35. 2.6 In your country, are there specific funding mechanisms[i] for open science at the national level? (ref.: (ii) 17.j, (iii) 18.j, (iv) 19.c)

[i] A funding mechanism refers to a policy instrument or process through which financial resources are allocated or provided for a specific purpose. Some examples are block funding, research grants, support for technological poles and centres of excellence, support for science parks and innovation centres, Infrastructure grants (for research facilities, labs, instruments), communication and outreach funds, innovation funds, loans and tax credits scholarships, studentships, fellowships, trust funds, sectoral funds.

- ☒ Yes
- ☐ Under development
- ☐ No

36. 2.6 cont. **If yes or under development**, please provide details and links as relevant.

17.i – Promoting the creation of equal public-private partnerships for open science and engaging the private sector in open science activities, subject to the availability of appropriate certification and regulatory instruments... 18.i – Public agreements reached within regional or global research communities and defining the community's data exchange methods, data formats, metadata standards, ontologies, terminology, tools and infrastructure... 19.c – Financing and supporting advanced educational programs and the professionalization of functions in the field of data science and data management... In terms of using the mechanisms for financing open science, it should be noted that several such mechanisms operate in Belarus: 1. BRFFR grants (example: https://fond.bas-net.by/if201_.html). A prerequisite for the provision of grants from the Belarusian Republican Foundation for Fundamental Research (BRFFR) is the commitment of scientists to make the results of their research public domain by publishing them in scientific journals with a mandatory indication of financial support from the BRFFR. 2. Instruments of state support for innovation projects at the expense of the Belarusian Innovation Fund. 2.1. Innovation voucher (<https://belinfund.by/deyatelnost/vauchery-i-granty/vauchery/>): is provided on a non-refundable basis; is issued to individuals and sole proprietors for the implementation of a project at the initial stage (preparatory or design and technological stage). 2.2. Grants (<https://belinfund.by/deyatelnost/vauchery-i-granty/granty/>): is provided on a non-refundable basis; issued to SMEs during the implementation of the design and technological stage for the implementation of scientific, technical, experimental design and technological work; financing is provided on the condition of financial participation of at least 10% of the total amount of necessary funds.

37. 2.7 1.1 In your country, have open science practices[i] been integrated in existing research funding mechanisms?

[i] Examples can include sharing open scientific knowledge (publications, research data, educational resources, software and hardware) with other scientists and with the public, sharing or using open infrastructures, collaboration with other researchers, with national and local societal actors such as policy makers, or sectors such as industry, agriculture, and health, and collaboration and open dialogue with indigenous peoples and local communities.

☒ Yes

☐ No

38. 2.7 cont. **If yes**, please provide details and links as relevant.

1. In this part, it should be noted that the country implements the joint use of open infrastructure: industry laboratories that are created to ensure the implementation of research, development and experimental-technological work, scientific support for innovative projects, pilot testing and the introduction of the results of scientific and scientific-technical activities into production in organizations in a specialized field or industry; centers for the collective use of unique scientific equipment (CCU). The CCU network is developing in order to increase the efficiency of using scientific equipment necessary for the development of priority areas of scientific and scientific-technical activities. 2. Close cooperation is being implemented with scientists within the framework of the Union State, the EAEU, the CIS, and international cooperation is developing with Asian countries. Thus, in 2024, Belarusian scientists implemented more than 90 joint scientific and technical projects with partners from seven countries, including Azerbaijan, Armenia, India, China, Moldova, Serbia and Uzbekistan. New competitions for joint projects were also announced, and competitions with Turkey, India, Tajikistan and Azerbaijan are planned for 2025. Within the Union State, by the Resolution of the Supreme State Council of the Union State dated January 29, 2024, No. 2, the Strategy for Scientific and Technological Development of the Union State for the period up to 2035 was approved (<https://soyuz.by/projects/dekrety-vysshego-gosudarstvennogo-soveta-soyuznogo-gosudarstva/postanovlenie-ot-29-yanvarya-2024-g-2-o-strategii-nauchno-tehnologicheskogo-razvitiya-soyuznogo-gosudarstva-na-period-do-2035-goda>), which was discussed both in the National Academy of Sciences of Belarus and in the Russian Academy of Sciences. The strategy provides for the formation of a single scientific, technological and educational space of the Union State. And one of the fundamental principles of scientific and technological development of the Union State is openness: effective interaction of organizations that perform R & D, participants in research and development with representatives of the business community, society and the state, as well as, based on national interests, with the international community. 3. As part of the cooperation between the scientific community and industry, state scientific and technical programs (SSTP) are implemented in the republic, the customers of which, among other things, are industry government bodies, for example (<https://www.gknt.gov.by/deyatelnost/gosudarstvennye-nauchno-tehnicheskie-programmy.php>).

39. 2.8 Have efforts been made or are foreseen to be undertaken by end of 2025 in your country to foster equitable public-private partnerships on open science in line with the 2021 Recommendation? (ref.: (ii) 17.i)

☒ Yes

☐ No

40. 2.8 cont. **If yes**, please provide details and links as relevant.

17.i – Promoting the creation of equal public-private partnerships in the interests of open science and involving the private sector in open science activities, subject to the availability of appropriate certification and regulatory instruments... The following mechanisms and instruments are used in the republic to develop public-private partnerships in the interests of open science: 1. Participation in the State Program for Innovative Development of the Republic of Belarus (SPID). Representatives of the private sector can submit their innovative projects for implementation to SPID and receive appropriate benefits for the purpose of implementing the relevant project. 2. Instruments of state support for innovative projects at the expense of the Belarusian Innovation Fund: innovation voucher (<https://belinfund.by/deyatelnost/vauchery-i-granty/vauchery/>); grant (<https://belinfund.by/deyatelnost/vauchery-i-granty/granty/>). 3. Participation in public councils: public advisory (expert) council for the development of entrepreneurship in the scientific, technical and innovative spheres under the State Committee on Science and Technology of the Republic of Belarus (Regulations on the Council: <https://www.gknt.gov.by/upload/Deyatelnost/Polozenie%20ob%20OKS.pdf>); interdepartmental supervisory council of scientific and educational networks of Belarus under the National Academy of Sciences of Belarus (Composition of the council: <https://nasb.gov.by/rus/docs/sovety/nauch-obr.pdf>); public advisory council for the development of entrepreneurship under the Ministry of Education of the Republic of Belarus (Regulations on the Council: <https://edu.gov.by/sistema-obrazovaniya/obshchestvennye-sovety/obshchestvennye-sovety/obshchestvenno-konsultativnyj-sovet-po-razvitiyu-predprinimatelstva/>). 4. Participation in working groups, round tables in the scientific substantiation of the development of state planning and forecasting documents: the National Strategy for Sustainable Development of the Republic of Belarus, the Program for Socio-Economic Development of the Republic of Belarus. Within the framework of this activity, the work of round tables and working groups is carried out.

41. 2.9 Is there a national monitoring framework for open science in your country? Or are there specific indicators on open science included in the national monitoring and evaluation framework for STI policy (ref.: (ii) 17.j, 23.c)

- ☐ Yes
- ☐ Under development
- ☒ No

Part 3. INVESTING IN OPEN SCIENCE INFRASTRUCTURES AND SERVICES

42. 3.1 What is the latest official data for your country's percentage of national gross domestic product (GDP) dedicated to research and development expenditure? Please also indicate the year of publication of this data, and to which year(s) it refers. (ref.: (iii) 18.a)

Data on the science intensity of the GDP of the Republic of Belarus are published annually. As of January 24, 2025, the latest official data on the science intensity of the GDP for 2023 is 0.58% (year of publication - 2024).

43. 3.2 What is the latest official data for the percentage of your country's population that have access to reliable internet and bandwidth? Please also indicate the year of publication of this data, and to which year(s) it refers. (ref.: (iii) 18.b)

Data on the population coverage by cellular mobile telecommunications services using LTE (4G) technology in the Republic of Belarus are published annually. As of January 24, 2025, the latest official data relate to 2023 - 98.4% (year of publication - 2024).

44. 3.3 Are there national research and education networks (NRENs)[i] active in the country? (ref.: (iii) 18.c)

[i] A national research and education network (NREN) is a specialised internet service provider dedicated to supporting the needs of the research and education communities within a country.

☒ Yes

☐ No

45. 3.3 cont. **If yes**, please provide more information and links as relevant, and indicate to which open science actors in your country those networks are accessible.

16.c – Promoting bibliodiversity through diversification of publication formats and media, including in the humanities and social sciences, and diversity of business models by supporting non-profit, academic and scientific community publishing models as a public good. In this part, it should be noted that the following are in operation in Belarus: 1. Network of technology parks. Today, there is a network of 14 technology parks in the country. Technology parks as subjects of innovation infrastructure occupy a key position in the development of innovative entrepreneurship in the Republic of Belarus. Technology parks actively promote the strengthening of integration processes between educational institutions and innovative enterprises in the production, personnel, and research spheres. 2. Network of industry laboratories. An industry laboratory is created for the purpose of ensuring the implementation of research, development and experimental-technological work, scientific support of innovative projects, pilot industrial testing and the introduction into production of the results of scientific and scientific-technical activity in organizations of a specialized field or industry. 3. Network of centers for the collective use of unique scientific equipment (CCU). The network of CCUs is being developed in order to increase the efficiency of the use of scientific equipment necessary for the development of priority areas of scientific and scientific-technical activity.

46. 3.4 Are there national or institutional open science infrastructures[i] in your country? (ref.: (iii) 18.d, 18.e, 18.k, 18.l, 18.m)

[i] Some examples of open science infrastructures include major shared scientific equipment or sets of instruments, open computational and data manipulation service infrastructures that enable collaborative and multidisciplinary data analysis, open science platforms and repositories for publications, research data and source codes, archives, open bibliometrics and scientometrics systems for assessing and analysing scientific domains, open laboratories, software forges and virtual research environments, open innovation testbeds, incubators, science museums, science parks and infrastructure for non-digital materials.

- ☒ Yes
- ☐ Under development
- ☐ No

47. 3.4 cont. **If yes or under development**, please indicate which type(s) of infrastructure:

- ☐ federated information technology infrastructure for open science, including high-performance computing, cloud computing and data storage
- ☐ community managed infrastructures, protocols and standards, for example those that support bibliodiversity and engagement with society
- ☐ platforms for exchanges and co-creation of knowledge between scientists and society
- ☐ community-based monitoring and information systems to complement national, regional and global data and information systems

48. 3.4 cont. **If yes or under development**, for each infrastructure, please provide links as relevant and more information with regard to their compliance with the 2021 Recommendation on Open Science in terms of inclusivity, accessibility, interoperability, community governance, FAIR (Findable, Accessible, Interoperable, and Reusable) and CARE (Collective Benefit, Authority to Control, Responsibility and Ethics) principles. Please indicate if the infrastructure is non-commercial.

18.d – Non-commercial infrastructure facilities, including computing facilities and public digital infrastructure and services, that support open science approaches... 18.e – Interconnected and diversified open science information technology infrastructure, including, where appropriate, high-performance computing, cloud computing, and data storage... 18.k – Infrastructure for non-digital materials (e.g., reagents). 18.l – Platforms for information sharing and co-creation of knowledge between scientists and society, including through predictable and sustainable funding for volunteer organizations conducting citizen science research with local participation. 18.m – Public monitoring and information systems that complement national, regional, and global data and information systems. The country has a national and institutional open science infrastructure, consisting of: 1. Networks of electronic libraries: BSUIR repository - an electronic archive of scientific publications of the staff and teachers of the Belarusian State University of Informatics and Radioelectronics. Provides access to educational and methodological developments, scientific publications, conference materials and much more. The interface is available in Russian (<https://libeldoc.bsuir.by/>); BGUKI repository - an electronic archive of scientific publications of the Belarusian State University of Culture and Arts. Provides access to publications of the staff and teachers of the university, university publications, the university library. The interface is available in Russian (<http://repository.buk.by/>); BNTU repository - the repository contains the full texts of methodological guidelines and methodological manuals, monographs, textbooks and teaching aids created at BNTU. The interface is available in Russian (<https://rep.bntu.by/>). In addition, all five scientific journals of BNTU are connected to the international system of bibliographic references Crossref for digital identification of objects; repository of Polesie State University - an electronic archive of collections of Polesie State University. Provides access to publications of university staff and teachers, university publications, and the university library. The interface is available in Russian (<https://rep.polessu.by/>); electronic archive of scientific periodicals of the National Academy of Sciences of Belarus - the electronic archive includes issues of periodicals from 2009 to 2016. Access to the full texts of journals can also be carried out in two ways: through the electronic catalog of the library and the list of Belarusian periodicals on the library website. The interface is available in Russian, Belarusian and English (<https://csl.bas-net.by/resursy/pereodika.asp>); BSU Electronic Library - the site provides access to the materials and collections of the Belarusian State University. The program interface is available in Russian and English. Some of the materials are available locally through the BSU network. The interface is available in Russian (<https://elib.bsu.by/>); Masherov VSU Repository - provides full-text access to dissertations and dissertation abstracts, materials of conferences, congresses, seminars, publications of the VSU scientific library, university periodicals, materials of the university staff and teachers. The interface is available in Russian (<https://rep.vsu.by/>); Infrastructure type: platforms for the exchange and joint creation of knowledge between scientists and society 2. The network of scientific and technical libraries, consisting of the Republican Scientific and Technical Library (RSTL) and its branches (<https://rlst.by/>). RSTL is a national depository of domestic and foreign literature on engineering, technology, industrial economics and related industries. It is a state repository of patent documentation of the Republic of Belarus, technical regulatory legal acts and other documents in the field of standardization, as well as industrial catalogs. Infrastructure type: pla

49. 3.5 Does your country host or fund any federated regional or international information technology infrastructure for open science? (ref.: (iii) 18.e, 18.f)

☐ Yes

☒ No

50. 3.5. cont. **If no**, are the existing open science infrastructures at regional and international levels accessible to all the relevant open science actors from your country?

☐ Yes

☒ No

☐ Other

51. 3.6 Is your country involved in any North-South, North-South-South and South-South collaborations to optimize infrastructure use and joint strategies for shared, multinational, regional and national open science platforms? (ref.: (iii) 18.g)

☒ Yes

☐ No

52. 3.6 cont. **If yes**, please provide links as relevant and more information, including the name of the infrastructure, date of establishment, language, target groups and beneficiaries, governance/community agreements, scope (e.g. research areas, elements of open science, stage of research, accessibility to all the relevant users in the region or internationally).

Close cooperation is being implemented with scientists within the Union State, the EAEU, the CIS, and international cooperation is developing with Asian countries. Thus, in 2024, Belarusian scientists implemented more than 90 joint scientific and technical projects with partners from seven countries, including Azerbaijan, Armenia, India, China, Moldova, Serbia, and Uzbekistan. New competitions for joint projects were also announced, and competitions with Turkey, India, Tajikistan, and Azerbaijan are planned for 2025.

Part 4. INVESTING IN HUMAN RESOURCES, TRAINING, EDUCATION, DIGITAL LITERACY AND CAPACITY BUILDING FOR OPEN SCIENCE

53. 4.1 Has systematic capacity building on open science taken place in higher education and/or research performing institutions in your country? (ref.: (iv) 19.a, 19.c)

☒ Yes

☐ No

54. 4.1 cont. **If yes**, please indicate if those capacity building initiatives are formal or informal and provide more information and links as relevant.

19.a – Provide systematic and continuous capacity building in open science concepts and practices, including a broad understanding of the guiding principles and core values of open science... 24 19.c – Fund and support advanced educational programs and the professionalization of functions in the field of data science and management... With regard to the development of open science capacity in higher education institutions, it is necessary to note the presence of: electronic libraries (see paragraphs 1.4 and 3.4); technology parks at universities (e.g. RUE "UNPC "Technolab", Scientific and Technological Park of the Belarusian National Technical University "Polytechnic", etc.); student science and technology laboratories; educational courses on improving literacy in the field of intellectual property, conducted by the National Center of Intellectual Property of the Republic of Belarus; National Children's Technopark.

55. 4.2 Have capacity building programmes for policy makers on how to integrate core values and principles of open science in STI policies and strategies taken place in your country?

☐ Yes

☒ No

56. 4.2 cont. **If no**, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.

57. 4.3 Has a framework of open science competencies been incorporated into higher education research skills curricula in your country, or is it foreseen by the end of 2025? (ref.: (iv) 19.b)

☒ Yes

☐ No

58. 4.3 cont. **If yes**, please provide more information and links as relevant.

19.b – Coordination of a set of competencies in the field of open science, linked to individual disciplines for researchers at different stages of their careers... In this part, it should be noted that the educational programs of the bachelor's degree provide for the study of the disciplines "Fundamentals of Intellectual Property Management" (<http://edoc.bseu.by:8080/handle/edoc/18546>), "Methodology of Scientific Research" (http://edoc.bseu.by:8080/bitstream/edoc/101002/1/Metodologiya_nauchnykh_issledovaniy.pdf).

59. 4.4 Are open educational resources[i] as defined in the 2019 Recommendation on Open Educational Resources, used as an instrument for open science capacity building in your country in line with the 2019 Recommendation mentioned above? (ref.: (iv) 19.d)

[i] Open Educational Resources are learning, teaching and research materials in any format and medium that reside in the public domain or are under copyright that have been released under an open license, that permit no-cost access, re-use, re-purpose, adaptation and redistribution by others (2019 Recommendation on Open Educational Resources)

☐ Yes

☐ Partly

☒ No

60. 4.5 Have there been any initiatives in your country to support science communication in line with open science value and principles with a view to the dissemination of scientific knowledge to researchers of different disciplines, decision-makers and the public at large? (ref.: (iv) 19.e)

☒ Yes

☐ No

61. 4.5 cont. If yes, please provide more information and links as relevant.

19.e – Supporting scientific communication regarding open science methods in order to disseminate scientific knowledge among scientists from other scientific disciplines, decision-makers and the general public... Close cooperation is being implemented with scientists within the Union State, the EAEU, the CIS and international cooperation is being developed with Asian countries. Thus, in 2024, Belarusian scientists implemented more than 90 joint scientific and technical projects with partners from seven countries, including Azerbaijan, Armenia, India, China, Moldova, Serbia and Uzbekistan. New calls for joint projects have also been announced, and calls with Turkey, India, Tajikistan and Azerbaijan are planned for 2025. Within the Union State, by the Resolution of the Supreme State Council of the Union State dated January 29, 2024, No. 2, the Strategy for Scientific and Technological Development of the Union State for the period up to 2035 was approved (<https://soyuz.by/projects/dekrety-vysshego-gosudarstvennogo-soveta-soyuznogo-gosudarstva/postanovlenie-ot-29-yanvarya-2024-g-2-o-strategii-nauchno-tehnologicheskogo-razvitiya-soyuznogo-gosudarstva-na-period-do-2035-goda>), which was discussed both in the National Academy of Sciences of Belarus and in the Russian Academy of Sciences. The strategy provides for the formation of a single scientific, technological and educational space of the Union State. And one of the fundamental principles of scientific and technological development of the Union State is openness: effective interaction of organizations that perform R&D, participants in research and development with representatives of the business community, society and the state, as well as, based on national interests, with the international community.

Part 5. FOSTERING A CULTURE OF OPEN SCIENCE AND ALIGNING INCENTIVES FOR OPEN SCIENCE

62. 5.1 Have there been any initiatives at the national or institutional level, e.g. by universities and/or research performing institutions, to review the existing research assessment and evaluation processes in order to align them with open science principles and values or are they foreseen by the end of 2025? (ref.: (v) 20.b, 20c, 20i)

☒ Yes

☐ No

63. 5.1 cont. **If yes**, please provide more information and links as relevant.

In this part, the Resolution of the State Committee on Science and Technology of the Republic of Belarus dated 20.04.2017 No. 9 (<https://etalonline.by/document/?regnum=u217e2989>) approved the Methodological recommendations for assessing the effectiveness of scientific, scientific-technical and innovative developments and their implementation. The Methodological recommendations are intended for use in planning and assessing the effectiveness of the implementation of: state scientific research programs (SPNI) and their individual tasks; scientific and technical programs (STP) and their individual tasks; activities for scientific support of state programs; innovative projects. When assigning additional payments for an academic degree and academic title, the researcher's scientific activity is analyzed: participation in the implementation of scientific work (topics, programs, projects), participation in scientific events (conferences, round tables, seminars, symposia), scientific publications, reviewing scientific work, membership in editorial boards of scientific publications and specialized councils, training of scientific personnel (master's students, postgraduate students, doctoral students, applicants), training of personnel in a foreign language, and others.

64. 5.2 Have specific clauses that recognize open science practices, including sharing, collaborating and engaging with other researchers and societal actors beyond the scientific community, or dialogue with other knowledge systems, been incorporated in the career evaluation and progression systems or are foreseen by the end of 2025? (ref.: (v) 20.a, 20.c, 20.d)

☒ Yes

☐ No

65. 5.2 cont. **If yes**, please provide details and links as relevant.

20.a – Bring together a wide range of different stakeholders, including research funding agencies, universities, research institutions, 27 publishers and editors, and learned societies representing different disciplines and countries... 20.c – Promote the development and implementation of assessment and evaluation systems that... 20.d – Ensure that open science skills are widely known and taken into account as criteria for recruitment and promotion in the scientific and academic field. In this part, the country has developed and implemented the Database of Gifted Youth (<https://www.giac.by/programmy/bd-odarennoy-molodezhi.php>) - a system of centralized recording, accumulation and systematization of information about pupils, students, cadets who have received incentives from the special fund of the President of the Republic of Belarus for social support of gifted pupils and students and who are winners of international and republican Olympiads (competitions, tournaments, other competitions) in academic subjects, who have shown a high level of training in individual subject areas, special successes in research, creative, intellectual activities. Pupils, students, cadets included in the database of gifted youth have a number of benefits and advantages (<https://edu.gov.by/molodezhnaya-politika/podderzhka-molodezhi/%D0%91%D0%B0%D0%BD%D0%BA%20%D0%B4%D0%B0%D0%BD%D0%BD%D1%8B%D1%85%20%D0%BE%D0%B4%D0%B0%D1%80%D0%B5%D0%BD%D0%BD%D0%BE%D0%B9%20%D0%BC%D0%BE%D0%BB%D0%BE%D0%B4%D0%B5%D0%B6%D0%B8%202.0.docx>): – according to paragraph 10 of Article 72 Code of the Republic of Belarus on Education, citizens who are graduates of state institutions of vocational, secondary specialized and higher education, included in the database of gifted youth, have the right to priority distribution; - heads of state bodies and other organizations ensure the creation of conditions for the professional growth of citizens, included in the database of gifted youth, by: assigning them supervision (patronage) from experienced specialists, teachers, scientists, outstanding cultural figures; referring them to internships, retraining, advanced training; involving them in teaching and research activities; involving them in competitive selection of citizens, included in the database of gifted and young people, when forming personnel reserves; providing the right to priority receipt of residential premises in a hostel for the period of study; providing other guarantees in accordance with the legislation; – according to the Decree of the President of the Republic of Belarus of January 6, 2012, No. 13 "On Certain Issues of Providing Citizens with State Support for the Construction (Reconstruction) or Acquisition of Residential Premises": the right to receive preferential loans for the construction (reconstruction) or acquisition of residential premises is granted to young adult citizens who are laureates of the special fund of the President of the Republic of Belarus for the social support of gifted pupils and students, - in agreement with the Ministry of Education, and in accordance with the documents confirming such title. In this Decree, young citizens are understood to mean persons under 31 years of age and others. In addition, researchers with academic degrees and academic titles are assigned additional payments in accordance with the Decree of the President of the Republic of Belarus dated September 25, 2007 No. 450 "On the establishment of allowances for academic degrees and titles" (<https://etalonline.by/document/?regnum=P30700450>).

66. 5.3 Have there been any initiatives in your country to recognize and reward open science practices or are they foreseen by the end of 2025? (ref.: (v) 20.a, 20.b, 20.c, 20.e, 20.f, 20.g, 20h)

☒ Yes

☐ No

67. 5.3 cont. **If yes**, for each initiative, please provide more information, including on the stakeholders[i] that are involved or lead the initiative, and links as relevant.

[i] For example, research funders, universities, research institutions, learned societies, scientific publishers and journal editorial boards.

20.a – Bring together a broad range of different stakeholders, including research funding agencies, universities, research institutions, publishers and editors, and learned societies from different disciplines and countries... 20.b – Review research and professional performance assessment systems to align them with open science principles... 20.c – Promote the development and implementation of assessment and evaluation systems that... 20.e – Encourage funding agencies, research institutions, journal editorial boards, learned societies, and publishers to pursue policies... 20.f – Ensure a diversity of forms of scholarly communication that respect the principles of open, transparent, and equitable access, and support not-for-profit and partnership publishing models that do not charge for articles and books. 20.g – Applying effective governance measures and appropriate legislative regulation to address the problem... 20.h – Promoting the dissemination of publicly available materials and developing existing schemes for open licenses and exceptions to copyright and other intellectual property rights... In this part, a number of exhibition events are held in the country, the purpose of which is to highlight the achievements of Belarusian science: exhibitions organized by the State Enterprise "BelExpo";

68. 5.4 Have there been any unintended negative consequences[i] of open science practices reported in your country? (ref.: (v) 20)

[i] For example, predatory behaviours, data migration, exploitation and privatization of research data, increased costs for scientists and high article processing charges associated with certain business models in scientific publishing that may be causes of inequality for the scientific communities around the world and, in some cases, the loss of intellectual property and knowledge.

☐ Yes

☒ No

Part 6. PROMOTING INNOVATIVE APPROACHES FOR OPEN SCIENCE AT DIFFERENT STAGES

OF THE SCIENTIFIC PROCESS

69. 6.1 Have there been any initiatives at the national or institutional level that promote innovative, participatory methodological or procedural changes at different stages of the research cycle to increase the openness of the entire scientific process? (ref.: (vi) 21)

☐ YES, at the national level

☒ YES, at the institutional level

☐ YES, at both the national and institutional levels

☐ No

70. 6.1 cont. **If yes**, please indicate which of the recommended actions they address:

- ☐ promotion of pre-prints
- ☐ exploration of open peer-review practices
- ☐ valuing and sharing of negative scientific results and associated data
- ☐ participatory methods that value inputs from societal actors, e.g. citizen science, crowdsourcing scientific projects
- ☐ development of participatory strategies for identifying the needs of marginalized communities and highlighting socially relevant issues to be incorporated into research agendas
- ☐ promoting open innovation practices
- ☐ Other

71. 6.1. cont. **If yes**, please provide more information and links as relevant

YES, at the institutional level. BSU, BNTU, and other higher education institutions are particularly active. Assistance is provided to the promotion of open science practices.
<http://philosophy.by/ru/news/event-2024-03-05/> <https://bgut.by/library/news/7-shagov-k-otkrytoy-nauke>

Part 7. PROMOTING INTERNATIONAL AND MULTI-STAKEHOLDER COOPERATION IN THE CONTEXT

OF OPEN SCIENCE AND WITH VIEW TO REDUCING DIGITAL, TECHNOLOGICAL AND KNOWLEDGE GAPS

72. 7.1 How is your country promoting and facilitating international scientific collaborations on open science as outlined in the 2021 Recommendation on Open Science? (ref.: (vii) 22.a)

73. 7.2 Does your country have a strategy or a plan for stimulating cross-border multi-stakeholder collaboration on open science, including with regards to exchange of best practices, capacity building and metrics for open science? (ref.: (vii) 22.b, 22.e, 22.f)

☒ Yes

☐ No

74. 7.2 cont. **If yes**, please provide links as relevant and more information.

Close cooperation is being implemented with scientists within the Union State, the EAEU, the CIS, and international cooperation is developing with Asian countries. Thus, in 2024, Belarusian scientists implemented more than 90 joint scientific and technical projects with partners from seven countries, including Azerbaijan, Armenia, India, China, Moldova, Serbia, and Uzbekistan. New competitions for joint projects were also announced, and competitions with Turkey, India, Tajikistan, and Azerbaijan are planned for 2025. Within the framework of the Union State, by the Resolution of the Supreme State Council of the Union State dated January 29, 2024, No. 2, the Strategy for Scientific and Technological Development of the Union State for the period up to 2035 was approved (<https://soyuz.by/projects/dekrety-vysshego-gosudarstvennogo-soveta-soyuznogo-gosudarstva/postanovlenie-ot-29-yanvarya-2024-g-2-o-strategii-nauchno-tehnologicheskogo-razvitiya-soyuznogo-gosudarstva-na-period-do-2035-goda>), which was discussed both in the National Academy of Sciences of Belarus and in the Russian Academy of Sciences. This Strategy provides for the formation of a single scientific, technological and educational space of the Union State. And one of the fundamental principles of scientific and technological development of the Union State is openness: effective interaction of organizations performing R&D, participants in research and development with representatives of the business community, society and the state, as well as based on national interests with the international community.

75. 7.3 Is your country involved in any discussion on the establishment of regional and international funding mechanisms for open science? (ref.: (vii) 22.c)

☒ Yes

☐ No

76. 7.3 cont. **If yes**, please provide links as relevant and more information.

In this regard, the Republic of Belarus participates in Union programs and projects financed from the budget of the Union State. Examples include the following Union programs: "SG Complex" - development of basic elements of orbital and ground-based facilities for future multi-satellite constellations of small-sized spacecraft for observing the Earth's surface and near-Earth space; "Intelauto" - development of an on-board electronics system for motor vehicles that are superior to existing analogues, including engine control, on-board safety, robotic control, highly efficient electric motors and other components for electric and hybrid transport; "Component-F" - development of the most popular photonic materials, components and modules for mechanical engineering and machine tool building, information technology and communications, thermonuclear energy and the aerospace industry, as well as special equipment.

Part 8. GENERAL CONSIDERATIONS

77. 8.1 Are there any specific external causes that may impede the implementation of the 2021 UNESCO Recommendation on Open Science in your country?

☐ Yes

☒ No

78. ANY OTHER COMMENTS: If you wish, you may enter any additional comments or information here. You may also contact openscience@unesco.org